

Background and 511 keV line sensitivity of a balloon-borne TES Laue-lens telescope^{*}

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Abstract

The Galactic 511 keV annihilation line is a direct tracer of low-energy positrons, but its compact-source content and narrow-line kinematics remain difficult to measure because MeV instruments are strongly background dominated. We present a full-chain Monte Carlo and rate-folding study for a balloon-borne pointed 511 keV telescope that combines a Ge(111) Laue lens with a transition-edge-sensor (TES) microcalorimeter focal plane. The model couples the focused optical phase space to an engineering detector/cryostat geometry, prompt atmospheric backgrounds, activation buildup, exact-position delayed-source transport, active-shield vetoes, Compton/field-of-view consistency cuts, a narrow line window, and mission-time rate variation. The current v3p5 full-stat exact-position chain uses a 10 m focal-length Laue configuration with $A_{\text{eff}}(511 \text{ keV}) = 20.08476 \text{ cm}^2$ and a 510.58–511.42 keV analysis window. At a reference point-source flux of $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$, the selected line-window background and signal rates are 0.0624651 s^{-1} and $0.00118117 \text{ s}^{-1}$, respectively. The mission-time fold gives $Z_{20\text{d}} = 6.15522$, a 3σ detection time of 4.73758 d, a 20-d 3σ flux threshold of $4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$, and a 1 Ms threshold of $6.32564 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$. A transport-backed exact-position support-size and seed sweep changes the total line-window background by only 1.119% and $Z_{20\text{d}}$ by 0.551%. We also report spatial-analysis and atmospheric-line-of-sight sidecars, while keeping the hard-window exact-position chain as the primary rate authority. The results indicate that focused TES spectroscopy can reach scientifically useful narrow-line sensitivity from a balloon platform, provided activation and prompt atmospheric backgrounds are modeled in the same detector-coupled selection space as the

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signal.

Keywords: Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, Monte Carlo background simulation, activation background

1. Introduction

The 511 keV gamma-ray line from electron–positron annihilation is one of the cleanest spectroscopic signatures of antimatter in the Galaxy. After positrons slow down in the interstellar medium, they annihilate either directly with electrons, producing two photons near 511 keV, or through positronium formation, producing the 511 keV line together with a three-photon continuum below the line. This radiative complex probes positron production, propagation, thermalization, and annihilation conditions [1]. The line was first reported from the Galactic-center direction by balloon-borne experiments and later identified unambiguously with high-resolution Ge detectors as positron-annihilation radiation [1]. The observed Galactic flux implies a positron annihilation rate of order $10^{43} \text{ e}^+ \text{ s}^{-1}$, making the origin of these positrons a long-standing problem in MeV gamma-ray astronomy [1, 4].

The candidate source classes span standard and non-standard physics. Conventional astrophysical candidates include β^+ decay of radioisotopes synthesized in massive stars, novae, core-collapse supernovae, and thermonuclear supernovae; pair creation in compact objects such as X-ray binaries, microquasars, pulsars, and the Galactic-center black hole; and cosmic-ray interactions [1, 2]. More speculative explanations include light dark matter decay or annihilation and hidden-sector scenarios [1, 3]. The difficulty is that the observed annihilation morphology is shaped not only by where positrons are produced, but also by how far they propagate before thermalization and annihilation. Spectroscopy constrains the annihilation medium through the line centroid, line width, and positronium fraction, while continuum limits above 511 keV constrain the injection energy [1, 2]. The most discriminating observables for source identification—compact-source content, small-scale structure, velocity shifts, and spatially resolved line broadening—remain limited by instrumental sensitivity, angular resolution, and background systematics.

The current observational picture has been shaped by OSSE/CGRO, INTEGRAL/SPI, and COSI. INTEGRAL/SPI established a bright bulge-like

component and fainter disk emission, with the detailed bulge/disk decomposition depending on the adopted sky model and instrumental-background treatment [2, 3]. Recent long-exposure SPI analyses recover the central, broad-bulge, and disk structures and suggest marginal additional structures associated with massive-star regions [3]. COSI provided an important independent measurement from a 46-d super-pressure-balloon flight, detecting the Galactic 511 keV signal with Compton imaging and finding that the emission is broader than a single point source [4, 5]. These results demonstrate both the scientific value and the practical difficulty of MeV line astronomy: the celestial line is weak, while atmospheric and instrumental backgrounds are strong and time dependent.

Wide-field Compton telescopes are essential for mapping the large-scale 511 keV sky. A complementary strategy is to use a pointed telescope that concentrates 511 keV photons onto a small, high-resolution focal plane. A pointed system is especially useful for testing whether a fraction of the annihilation luminosity is produced by unresolved point sources, a bright central core, or transient pair-plasma episodes in compact objects. This motivation is reinforced by annihilation signatures reported during the 2015 outburst of V404 Cygni, which indicate that accreting stellar-mass black holes can produce pair plasma at astrophysically relevant rates [6]. Such a payload does not replace wide-field Compton missions; it addresses a different observable in a narrower field of view.

The 511-CAM concept follows this route by combining gamma-ray concentrating optics with stacked transition-edge-sensor microcalorimeter arrays [7]. It is attractive because it joins photon concentration with sub-keV calorimetric energy resolution, two capabilities rarely available together at MeV energies. A projected 511 keV detector-assembly resolution of a few 10^2 eV FWHM corresponds to a velocity scale of a few 10^2 km s $^{-1}$ and could enable velocity-shift or line-broadening measurements in favorable cases [7]. Event topology also matters. Single-pixel photoelectric events provide the cleanest spectroscopy, while multi-pixel events can be checked against Compton kinematics and rejected if inconsistent with photons arriving through the focusing optics.

The central technical challenge is background. Balloon-borne gamma-ray spectrometers operate in a strong radiation environment and are necessarily background dominated. Classical studies of balloon-borne Ge spectrometers identify atmospheric-neutron scattering, atmospheric and cosmic gamma-ray aperture flux, beta decays of activation products, and shield leakage as major

background components; they also emphasize minimizing passive material near the detector and lowering effective shield thresholds where possible [8]. For a 511 keV line payload, the problem is sharper because the science signal lies at the same energy as strong atmospheric and instrumental annihilation features. A credible sensitivity estimate therefore requires a detector-coupled model of the optics, entrance windows, cryostat, passive material, active shield, prompt atmospheric components, and delayed activation.

Detailed Monte Carlo mass modeling is the standard route to such estimates. The INTEGRAL/SPI response production required a high-fidelity mass model, ground- and in-flight-calibration comparisons, and a practical separation between imaging response and redistribution response to control computing and storage costs [9]. Modern MeV mission studies, including AMEGO and COSI, similarly rely on Geant4/MEGAlib-based simulations to calculate response, background, event reconstruction, and sensitivity [10–14]. The same philosophy is adopted here, but applied to a narrower and more demanding target: a balloon-borne, focused, narrow-line 511 keV telescope with a TES focal plane.

This paper presents a detector-coupled simulation and sensitivity study for such a payload. The simulation chain starts from a Ge(111) Laue-lens source model at 511 keV and transports focused photons through the detector entrance window into an engineering detector/cryostat mass model. The modeled assembly includes the TES focal plane, staged thermal windows, a Be entrance window, Cu/W passive shielding, cryogenic service proxies, and an active CsI shield. The background chain includes prompt atmospheric components, activation buildup, exact-position delayed-source transport, active-shield vetoes, Compton/field-of-view consistency cuts, a narrow 511 keV line window, mission-time rate evolution, and counting-significance estimates. The main contribution is not an isolated effective-area estimate, but a closed rate calculation in which signal and background pass through the same detector-coupled selection definitions.

The remainder of the paper is organized as follows. Section 2 summarizes the science-driven instrument requirements. Section 3 describes the simulation chain. Sections 4–7 describe the detector geometry, optics bridge, background model, and event selections. Section 8 presents the sensitivity, convergence, and background-composition results. Section 9 discusses interpretation, limitations, and future work. Section 10 summarizes the main conclusions.

Placeholder for workflow diagram: geometry → prompt/buildup
 transport → exact-position delayed source → Laue optics →
 EventList detector replay → veto/Compton/FoV selection → time
 fold → significance.

Figure 1: Conceptual simulation chain used in this work. The important feature is that the focused signal and all background streams are evaluated in the same detector-response and event-selection space.

2. Science and instrument requirements

The instrument considered here is optimized for a pointed, narrow-line measurement rather than all-sky imaging. The primary science case is the detection or exclusion of compact 511 keV emission at fluxes of order $10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$ or below during a long-duration balloon exposure. This flux scale is motivated by the need to test bright central-core, compact-source, and transient pair-plasma scenarios while remaining complementary to wide-field Compton surveys.

Four requirements follow. First, the telescope must concentrate 511 keV photons so that a small focal plane can be used without sacrificing the collecting area. Second, the focal plane must provide sub-keV energy resolution so that the line-window background scales with a narrow spectral interval rather than a broad MeV-band continuum. Third, the detector and shield design must suppress atmospheric, activation, and side-entry backgrounds without placing excessive passive material near the TES absorbers. Fourth, the simulation chain must treat prompt background, delayed activation, focused signal, vetoes, event topology, and mission-time variation consistently. These requirements define the methods used below.

3. Simulation workflow

Figure 1 summarizes the workflow. The calculation is divided into geometry definition, prompt and buildup transport, delayed-source construction, focused-optics transport, detector-coupled signal replay, detector event selection, mission-time rate folding, source-case normalization, and cumulative significance evaluation. The present manuscript uses the v3p5 full-stat exact-position branch as the primary rate authority. The conservative radial-profile delayed-source branch is retained only as a cross-check.

133 The rate-level authority is the exact-position delayed-source chain labeled
 134 `fullstat_v2_exactpos`. It uses full-stat prompt and buildup transport, a
 135 fixed day-15 delayed inventory, sampled exact-position `PointSource` delayed
 136 support, and full downstream propagation through the Step05–Step08 de-
 137 tector response and significance chain. The older homogeneous-beam and
 138 legacy delayed-source branches are not used for the headline sensitivities.

139 4. Detector and cryostat mass model

140 The detector model is an engineering Geant4/MEGAlib proxy for a balloon-
 141 borne TES focal-plane assembly. It is not intended to be a final CAD repro-
 142 duction; rather, it captures the material and geometrical features most rel-
 143 evant to MeV background formation and event selection. The current v3p5
 144 center-finger geometry contains a TES pixel array, a Be entrance window,
 145 staged Al thermal windows, Cu/W passive shields and collimation, cryogenic
 146 service-mass proxies, and a segmented CsI active shield. The local side-entry
 147 geometry points the window 45° upward in the zenith-frame source coordi-
 148 nates, with an instrument-frame rotation of 0, 45, 0 degrees. A 2 cm-thick W
 149 square-bore side collimator defines the local side-window entrance.

150 The active shield is treated quantitatively by post-processing rather than
 151 by relying on native production-trigger filtering. Candidate events are grouped
 152 on a common Poisson time axis and vetoed if the summed CsI shield energy
 153 in the coincidence window exceeds the analysis threshold. This choice avoids
 154 filtering activation-production storage at the transport stage and keeps the
 155 veto definition common to prompt, delayed, and focused-science streams.

156 5. Laue optics and detector-coupled signal replay

157 The signal model uses a Ge(111) Laue-lens configuration tuned to 511
 158 keV. The current f10m A1 optical authority has focal length 10 m, 25 tiles, 18
 159 mm tile size, 30 arcsec mosaicity, active Ge mass 0.44063 kg, and $A_{\text{eff}}(511 \text{ keV}) =$
 160 20.08476 cm^2 . A three-seed B-FULL focal-crossing aggregate contains 37,194
 161 within-Be focal rows, with within-Be fraction 0.998336, $r_{90} = 1.02853 \text{ cm}$,
 162 and $r_{99} = 1.24527 \text{ cm}$.

163 The optics and detector are coupled through an EventList bridge. Focal-
 164 crossing rows accepted inside the Be-window gate are transformed into the lo-
 165 cal detector coordinate system and replayed through the full detector/cryostat
 166 geometry. This avoids using an analytic phase-space projection as the signal

Table 1: Key detector and analysis parameters used for the primary v3p5 exact-position chain.

Parameter	Value or policy
Detector concept	TES microcalorimeter focal plane
Active shield	Segmented CsI, post-processing veto
Entrance window	Be side window
Passive shielding/collimation	Cu/W proxy shields and 2 cm W side sleeve
Pointing	Local side window tilted 45° upward
Detector trigger/veto authority	Step05 post-processing selection
Line window W_2	510.58–511.42 keV
Reference flux F_{ref}	$10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$

authority and preserves the optical focusing handoff at the detector entrance plane. The v3p5 bridge transported 37,194/37,194 EventList rows. At $F_{\text{ref}} = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$, the Step05 hard-window signal response is $0.00118117 \text{ s}^{-1}$, corresponding to a Step07 response of $11.8117 \text{ cps}/(\text{ph cm}^{-2} \text{ s}^{-1})$.

6. Prompt and delayed background model

The background model contains prompt atmospheric transport and delayed activation. The full-stat prompt stream and the activation-buildup stream each generated 25,210,216 particles. Prompt normalization uses per-tag exposure normalization, while delayed activation is constructed from the activation-production ledgers and corrected ground-state half-lives. The fixed day-15 delayed inventory activity for the current chain is 85.636696 Bq.

The main methodological upgrade in the current branch is exact-position delayed-source transport. Instead of compressing production positions into axisymmetric radial-profile source blocks, the exact-position chain samples equal-flux `PointSource` support blocks from weighted radioactive-production rows. The baseline exact-position source uses $M = 5000$ support blocks with seed 260613, contains no `RadialProfileBeam` blocks, and preserves the fixed activity budget. The delayed transport used 10^6 triggers and stored 10^6 events, with observation time 11530.473845 s.

This exact-position source representation is not a one-block-per-production-row stress test. Instead, it is a statistically sampled representation whose adequacy is evaluated downstream in Section 8.3. The convergence test is

189 performed on the final selected line-window rates and significance, not only
 190 on source-sampling diagnostics.

191 7. Event selection and significance calculation

192 The analysis window is

$$W2 = [510.58, 511.42] \text{ keV}, \quad (1)$$

193 corresponding to $511 \text{ keV} \pm 420 \text{ eV}$. This window is matched to a sub-keV
 194 TES response assumption and is narrow enough to suppress continuum back-
 195 ground while retaining most of the line response.

196 Events are processed on a common time axis. Prompt, delayed, and fo-
 197 cused signal event instances are drawn according to their rates, sorted by
 198 time, and grouped within a $1 \mu\text{s}$ coincidence window. The active-shield veto
 199 is then applied to the summed shield energy. TES hit topologies are classi-
 200 fied with a custom Compton/field-of-view consistency test. One-hit events
 201 are retained. Two-hit events test both possible first-scatter orders. Higher-
 202 multiplicity events enumerate candidate orders and use a Compton sequence
 203 quality metric before testing whether the first-scatter cone is consistent with
 204 the Be-window acceptance. Events without a valid reconstruction are re-
 205 tained under the current conservative rejection policy.

206 The time-dependent significance is computed as a cumulative counting
 207 statistic. For a reference flux F_{ref} , the cumulative significance is

$$Z(t) = \frac{N_s(t)}{\sqrt{N_b(t)}}, \quad (2)$$

208 where $N_s(t)$ and $N_b(t)$ are the time-integrated selected signal and background
 209 counts after mission-time scaling and live-factor corrections. For comparison
 210 among analysis variants, the 3σ flux threshold is scaled as

$$F_{3\sigma}(t) = F_{\text{ref}} \frac{3}{Z(t)}. \quad (3)$$

211 This Gaussian counting convention is used consistently for the hard-window
 212 results and sidecar comparisons.

Table 2: Primary sensitivity and selected comparison cases. The exact-position hard-window row is the current rate authority. The radial-profile row is a conservative baseline cross-check. Spatial and line-of-sight rows are sidecars, not replacements for the hard-window authority.

Case	B_{W2} (cps)	S_{W2} at F_{ref} (cps)	$Z_{20\text{d}}$	$T_{3\sigma}$ (d)	$F_{3\sigma}(20\text{d})$
Exact-position hard window	0.0624651	0.00118117	6.15522	4.73758	4.87391×10^{-5}
Radial-profile baseline	0.0729576	0.00118117	5.70221	5.46687	5.26111×10^{-5}
Focused r_{90} sidecar	0.0232510	≈ 0.001063	8.17566	2.23643	3.66943×10^{-5}
45° LOS sidecar	—	—	5.02544	—	5.96962×10^{-5}
Multi-annulus sidecar	—	—	8.45804	—	3.54692×10^{-5}

213 8. Results

214 8.1. Primary hard-window sensitivity

215 Table 2 gives the primary hard-window result. In the exact-position chain,
 216 the Step05 selected $W2$ background and signal rates at $F_{\text{ref}} = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$
 217 are 0.0624651 s^{-1} and $0.00118117 \text{ s}^{-1}$, respectively. After the mission-time
 218 fold, the mean background and signal rates are 0.0626835 s^{-1} and $0.00117281 \text{ s}^{-1}$.
 219 The resulting 20-d significance is $Z_{20\text{d}} = 6.15522$, with a 3σ detection time
 220 of 4.73758 d. The 20-d 3σ flux threshold is $4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$, and the
 221 1 Ms threshold is $6.32564 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$.

222 The hard-window result is intentionally conservative relative to the spatial
 223 sidecars. It requires only the narrow energy window, active-shield veto, and
 224 topology/FoV consistency selection. The r_{90} sidecar illustrates the possible
 225 gain from using the focused spot in the local side-window detector frame,
 226 but it should be treated as a post-processing spatial analysis rather than a
 227 finalized profile-likelihood result.

228 8.2. Background composition

229 The selected exact-position $W2$ background is dominated by prompt at-
 230 mospheric events. The Step05 stream split is 0.0590827 s^{-1} prompt and
 231 $0.00338234 \text{ s}^{-1}$ delayed, corresponding to 94.6% and 5.4% of the total selected
 232 hard-window background. The largest prompt component is the positron-
 233 tagged atmospheric stream, contributing 0.0543377 s^{-1} or 86.99% of the total
 234 $W2$ background. The delayed component is led by ^{64}Cu , which contributes
 235 $0.00251507 \text{ s}^{-1}$, or 74.36% of the delayed $W2$ events.

Table 3: Selected $W2$ background composition in the exact-position chain. Percentages are relative to the total $W2$ background unless otherwise stated.

Component	Rate (cps)	Fraction
All prompt atmospheric streams	0.0590827	94.6%
All delayed activation streams	0.00338234	5.4%
Prompt e^+ stream	0.0543377	86.99%
Delayed ^{64}Cu	0.00251507	74.36% of delayed

Table 4: Transport-backed exact-position convergence cases.

Label	M	Seed	B_{W2} (cps)	Z_{20d}
fullstat_v2_exactpos	5000	260613	0.0624651	6.15522
m05000_s260614	5000	260614	0.0631683	6.12141
m20000_s260613	20000	260613	0.0627261	6.14261
m50000_s260613	50000	260613	0.0629804	6.13039

236 This composition has two implications. First, improving the hard-window
 237 sensitivity is primarily a prompt-background problem for the current geom-
 238 etry and selection. Second, delayed activation is small but not negligible;
 239 its line-window contribution is sufficiently structured that exact-position de-
 240 layed transport is worth preserving in the chain.

241 8.3. Exact-position support-size and seed convergence

242 Table 4 summarizes the transport-backed exact-position convergence sweep.
 243 Four downstream cases were run: two seeds at $M = 5000$ and two larger sup-
 244 port sizes at $M = 20000$ and $M = 50000$. Across these cases, the delayed
 245 subcomponent shows visible Monte Carlo variation, but the total selected
 246 $W2$ background changes by only 1.119%, and Z_{20d} changes by only 0.551%.
 247 The final sensitivity is therefore stable at the level required for the present
 248 rate claim.

249 8.4. Comparison with instrument benchmarks

250 The present calculation is not an all-sky imaging forecast. It is a pointed,
 251 narrow-line, compact-source sensitivity estimate. The comparison most rel-
 252 evant to its design philosophy is therefore not simply field-of-view survey
 253 speed, but line-window background suppression per focused point-source flux.

254 INTEGRAL/SPI and COSI remain the key references for mapping the dif-
 255 fuse 511 keV sky [2–5]. COSI’s satellite implementation will survey the 0.2–5
 256 MeV sky with wide field of view and Ge detector spectroscopy, and public
 257 NASA mission information lists a planned 2027 launch and a 6.0 keV energy
 258 resolution at 511 keV [15, 16]. The TES Laue-lens concept considered here
 259 is complementary: it trades sky coverage for a concentrated beam and a
 260 sub-keV line window.

261 Relative to a wide-field Compton telescope, the proposed payload would
 262 be most competitive for compact 511 keV targets, central-core searches, and
 263 transient follow-up observations where the source position is known or con-
 264 strained. It is less suited to measuring the full diffuse bulge and disk lumi-
 265 nosity, because the small focused aperture samples only a small solid angle.
 266 This distinction is essential when comparing point-source Route-A results
 267 with diffuse-sky Route-B foreground estimates.

268 9. Discussion

269 The main result is that a focused TES line spectrometer can reach a
 270 20-d 3σ point-source threshold below $5 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$ in the current hard-
 271 window exact-position chain. This threshold is obtained without invoking
 272 the focused-spot spatial sidecar as the headline result. It follows from three
 273 combined effects: optical concentration, a sub-keV energy window, and a
 274 detector-coupled veto/topology selection that suppresses prompt and delayed
 275 backgrounds in the same selection space as the signal.

276 The exact-position delayed-source treatment is a methodological improve-
 277 ment over radial-profile compression. Although delayed activation contributes
 278 only about 5.4% of the selected hard-window background in the present chain,
 279 it is spatially and materially structured. Preserving sampled production po-
 280 sitions prevents artificial azimuthal or radial redistribution from becoming
 281 an unquantified systematic. The downstream convergence sweep indicates
 282 that $M = 5000$ support blocks are sufficient for the current hard-window
 283 rate claim, while larger support-size stress tests remain useful engineering
 284 checks.

285 The background composition points to the next optimization targets. The
 286 selected hard-window rate is dominated by prompt atmospheric e^+ -tagged
 287 events. Improvements may therefore come from shield geometry, side-entry
 288 collimation, material reduction near the focal plane, stricter topology cuts, or
 289 improved reconstruction of multi-pixel events. Delayed activation is smaller

but led by identifiable nuclides and volumes, suggesting targeted material substitutions or local shielding may still be valuable.

Several limitations should be kept explicit. First, the detector/cryostat model is an engineering proxy, not a final flight CAD model. Second, the production SIM files use a post-processing active-shield authority; native trigger/veto production filtering is not the source of quantitative rates. Third, the Step06 mission-time calculation is a rate-level fold, not a new particle transport at each trajectory time bin. Fourth, the custom Compton/FoV veto is not yet a full Revan/Mimrec reconstruction. Fifth, upstream Laue-lens self-background and activation from cosmic-ray interactions in the optics hardware are not included in the headline detector-background budget. Sixth, spatial sidecars such as r_{90} and multi-annulus likelihood estimates are promising, but they are not yet a publication-grade nuisance-profile likelihood.

These limitations do not invalidate the primary conclusion, but they define the next work required before a final instrument proposal or flight performance paper. The most valuable follow-up studies are a Revan cross-check of the multi-hit subset, an optics self-background simulation, a final cryostat/DR geometry update, measured or literature-anchored TES saturation and pile-up studies at the relevant rates, and a spatial-spectral likelihood that profiles background nuisance parameters.

10. Conclusions

We have presented a first full-chain detector-coupled background and sensitivity study for a balloon-borne pointed 511 keV TES Laue-lens telescope. The current v3p5 exact-position chain couples a Ge(111) Laue optical response to a TES focal-plane mass model, prompt atmospheric transport, delayed activation, active-shield vetoing, Compton/FoV topology selection, mission-time rate folding, and counting-significance evaluation. In the primary 510.58–511.42 keV hard window, the chain predicts $B = 0.0624651 \text{ s}^{-1}$ and $S = 0.00118117 \text{ s}^{-1}$ at $10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$, giving $Z_{20d} = 6.15522$ and $F_{3\sigma}(20d) = 4.87391 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$. Exact-position delayed-source convergence is stable at the final hard-window significance level. The result supports the feasibility of focused TES spectroscopy as a complementary 511 keV technique for compact-source and central-core searches, while identifying prompt atmospheric background suppression and reconstruction validation as the most important next steps.

326 **CRedit authorship contribution statement**

327 Placeholder: to be completed with the final author list.

328 **Declaration of competing interest**

329 The authors declare no competing financial interests. This statement
330 should be verified by all authors before submission.

331 **Data and software availability**

332 Simulation cards, validation summaries, and analysis scripts will be made
333 available in a public repository or institutional archive before publication.
334 Internal path names in this draft are placeholders and should be replaced by
335 persistent identifiers.

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